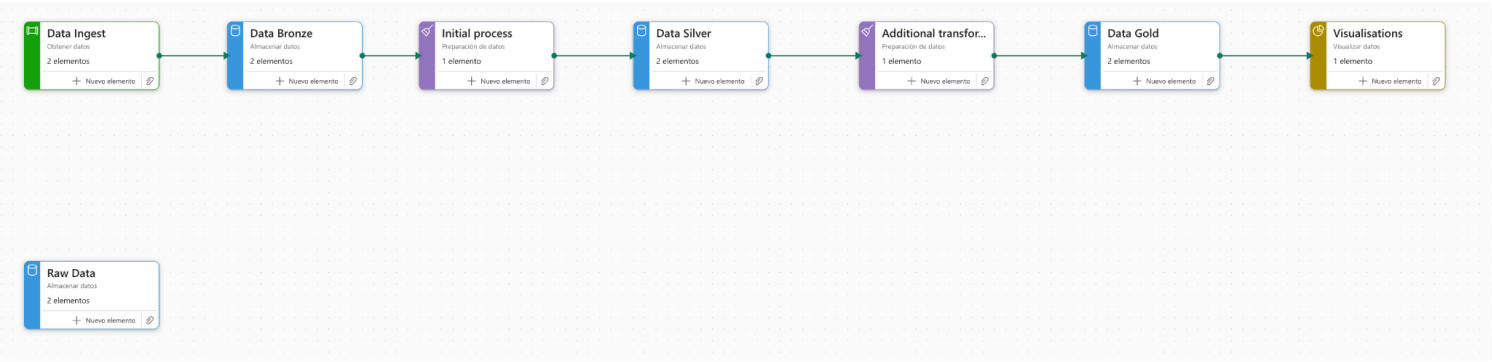
ETL Pipeline and Power BI Analytics Report – Amazon Sales Dataset

Dataset and Ingestion

Dataset Source: https://www.kaggle.com/datasets/rohiteng/amazon-sales-dataset

For this project, I selected a synthetic Amazon-style e-commerce sales dataset from Kaggle, containing 100,000 transactions with 20 well-structured columns. The dataset includes information about orders, customers, products, pricing, payments, logistics, and order outcomes. Although artificially generated, it reflects realistic online retail behavior, including dynamic pricing, discounts, multiple product categories and brands, seasonal trends, diverse payment methods, and realistic customer locations and names.



The raw dataset was first ingested into the Lakehouse Raw layer to simulate data collection from the source. Using a copy job, the data was moved to the Bronze layer for initial storage.

Data Cleaning and Transformation

The **silver_notebook** performs the following transformations to prepare the data for analytics:

- Remove alphabetical prefixes from IDs (OrderID, CustomerID, ProductID, SellerID).

- Cast numeric and financial columns to appropriate types (Integer and Decimal).

- Trim text fields (CustomerName, ProductName, Category, Brand, PaymentMethod, City, State, Country).

- Convert OrderDate to proper Date type.

- Check for duplicate OrderIDs.

- Save the cleaned DataFrame as a Delta table in the Silver layer for downstream processing.

The **gold_notebook** transforms the Silver dataset into a normalized star schema for Power BI. Key steps include:

Load Silver Data:

- Load the cleaned Silver dataset into a Spark DataFrame.

- This dataset has all IDs, numeric fields, and dates standardized, ready for transformation into fact and dimension tables.

Customer Dimension (dim_customers):

- Group records by CustomerID and retain the first occurrence of CustomerName.

- Ensures unique customers and provides a reference table for the fact table.

- Quality checks performed: total_rows counts all rows / distinct_ids counts unique CustomerID values, detecting potential duplicates.

Payment Method Dimension (dim_payment_methods):

- Select distinct PaymentMethod values.
- Generate unique PaymentMethodID for each using a row number over an ordered window.
- Row numbers are left-padded to 3 digits (e.g., "001"), creating a clean, consistent dimension table.

City Dimension Mapping

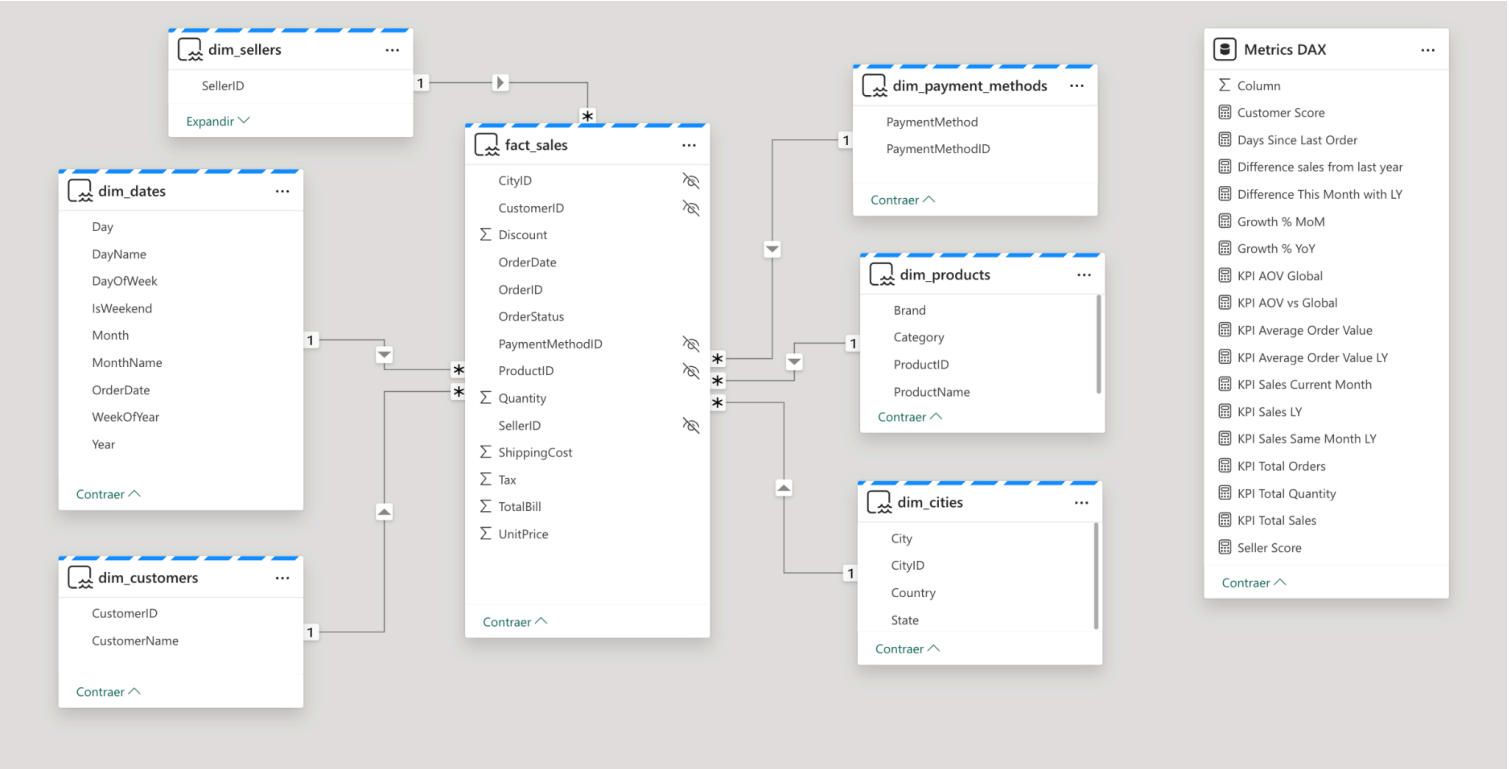
- Assign CityID to the fact table.
- Create a sequential number for each row using a window with a monotonically increasing ID.
- Map numbers to CityID values from 1 to 100 cyclically (modulo 100).
- Convert to zero-padded strings (e.g., "001") and drop temporary columns.
- Ensures each fact record references a valid city from dim_cities.

Fact Table (fact_sales)

- Calculate TotalBill = (UnitPrice * Quantity) + Tax + ShippingCost - Discount.
- Drop redundant descriptive columns now represented in dimensions:
- TotalAmount, CustomerName, ProductName, Category, Brand, PaymentMethod, City, State, Country.
- Keep only foreign keys and numeric measures for aggregation and KPI calculations.
- Save the fact table and all dimensions to the warehouse_gold.DataSets schema in Delta format, with overwrite and overwriteSchema=True to ensure tables are up-to-date.

This completes the ETL pipeline from Silver to Gold, producing a clean star schema ready for Power BI reporting.

Star Schema Model



The Gold layer implements a star schema:

Fact Table: fact_sales containing transactional metrics (TotalBill, Quantity, etc.) and foreign keys to dimensions.

Dimensions: Customers, Products, Dates, Payment Methods, Cities, Sellers.

Relationships are defined by IDs linking fact and dimension tables, ensuring efficient aggregation and analysis.

DAX Measures

The Power BI semantic model includes a variety of DAX measures designed to analyze sales performance, customer behavior, and seller performance:

Sales Metrics:

- Total sales, total orders, and total quantity, as fundamental aggregation metrics.

- Average Order Value (AOV) and AOV for previous year to track customer spending patterns.

- Month-to-date sales and same month last year comparison.

- Year-over-year growth and month-over-month growth percentages for trend analysis.

- Differences in sales compared to last year, both annually and monthly.

Customer and Seller Scoring:

- Customer Score and Seller Score combine normalized measures of quantity, number of orders, and total amount spent to create a single performance index.

- Days Since Last Order tracks customer recency, useful for identifying engagement levels.

Global Measures:

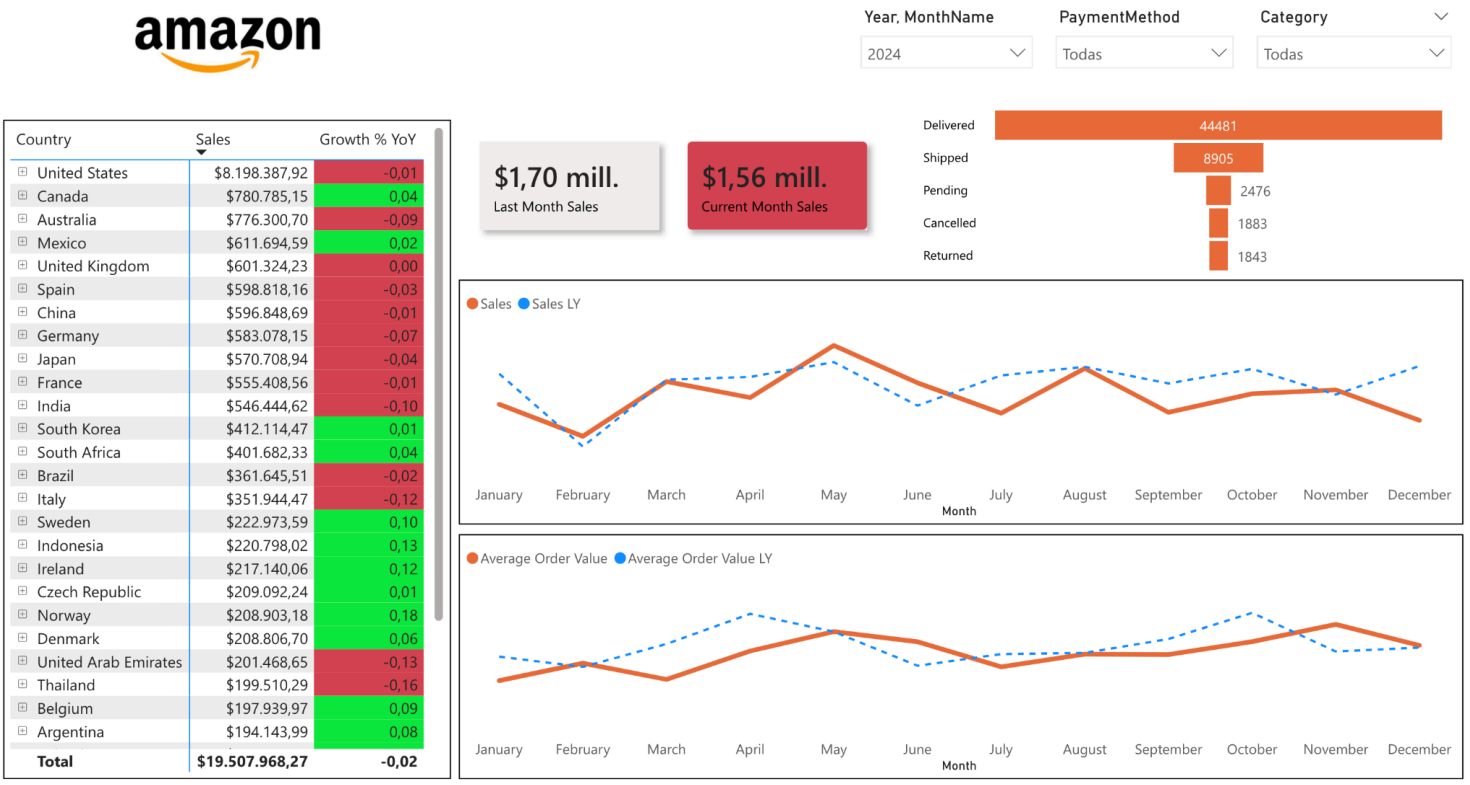
- Global average order value ignoring filters, useful for benchmarking performance across all customers and sellers.

These measures allow the reports to highlight top-performing products, customers, and sellers, as well as flag underperforming areas. Conditional formatting is applied to visualize positive vs negative trends (e.g., green for above-average performance, red for below).

Reports and Interactivity

Two Power BI reports were created, each designed to provide actionable insights with interactive elements:

Sales Report:



Country	Sales	Growth % YoY
United States	$8.198.387,92	-0,01
Canada	$780.785,15	0,04
Australia	$776.300,70	-0,09
Mexico	$611.694,59	0,02
United Kingdom	$601.324,23	0,00
Spain	$598.818,16	-0,03
China	$596.848,69	-0,01
Germany	$583.078,15	-0,07
Japan	$570.708,94	-0,04
France	$555.408,56	-0,01
India	$546.444,62	-0,10
South Korea	$412.114,47	0,01
South Africa	$401.682,33	0,04
Brazil	$361.645,51	-0,02
Italy	$351.944,47	-0,12
Sweden	$222.973,59	0,10
Indonesia	$220.798,02	0,13
Ireland	$217.140,06	0,12
Czech Republic	$209.092,24	0,01
Norway	$208.903,18	0,18
Denmark	$208.806,70	0,06
United Arab Emirates	$201.468,65	-0,13
Thailand	$199.510,29	-0,16
Belgium	$197.939,97	0,09
Argentina	$194.143,99	0,08
Total	**$19.507.968,27**	**-0,02**

Provides a high-level overview of sales performance, highlighting key metrics such as total sales, average order value (AOV), annual sales trends, and growth percentages.

Visualizes positive trends in green and negative trends in red, making it easy to spot profitable vs underperforming periods.

Includes charts showing annual sales and average order value by country, with the ability to drill down to regions and cities for deeper insights.

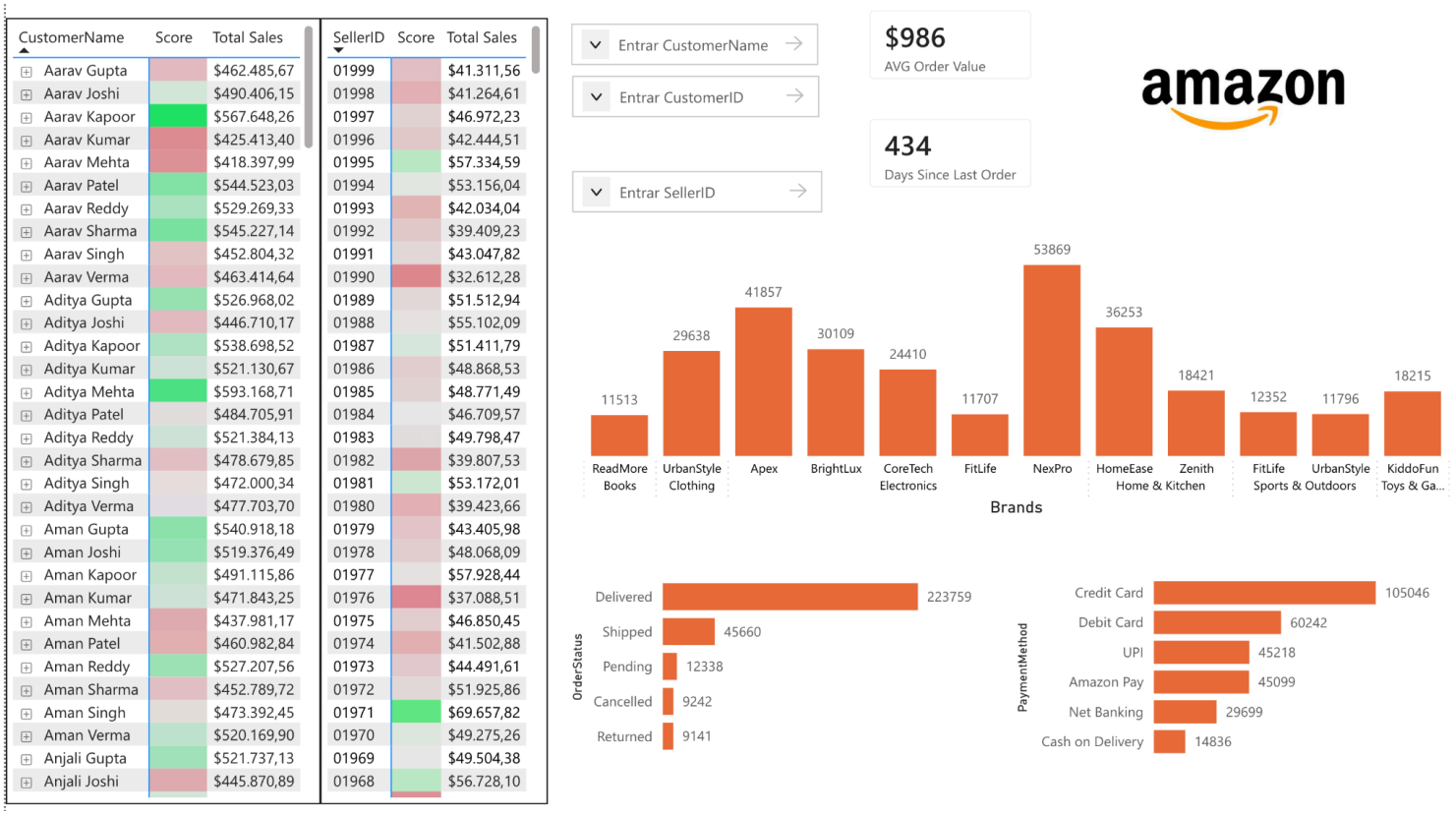
Order status is highlighted with an index, showing the proportion of delivered, returned, cancelled, or pending orders, helping detect potential issues in specific segments.

Interactive tables: Selecting rows in the summary tables automatically filters all related visuals across the report. For example, selecting a specific country, product,

or year updates the charts and KPIs dynamically, allowing users to analyze trends for that segment in real time.

Interactive filtering by product, year, month, and payment method allows users to dynamically explore the data and understand performance drivers.

Customer Report:



CustomerName	Score	Total Sales
Aarav Gupta		$462.485,67
Aarav Joshi		$490.406,15
Aarav Kapoor		$567.648,26
Aarav Kumar		$425.413,40
Aarav Mehta		$418.397,99
Aarav Patel		$544.523,03
Aarav Reddy		$529.269,33
Aarav Sharma		$545.227,14
Aarav Singh		$452.804,32
Aarav Verma		$463.414,64
Aditya Gupta		$526.968,02
Aditya Joshi		$446.710,17
Aditya Kapoor		$538.698,52
Aditya Kumar		$521.130,67
Aditya Mehta		$593.168,71
Aditya Patel		$484.705,91
Aditya Reddy		$521.384,13
Aditya Sharma		$478.679,85
Aditya Singh		$472.000,34
Aditya Verma		$477.703,70
Aman Gupta		$540.918,18
Aman Joshi		$519.376,49
Aman Kapoor		$491.115,86
Aman Kumar		$471.843,25
Aman Mehta		$437.981,17
Aman Patel		$460.982,84
Aman Reddy		$527.207,56
Aman Sharma		$452.789,72
Aman Singh		$473.392,45
Aman Verma		$520.169,90
Anjali Gupta		$521.737,13
Anjali Joshi		$445.870,89

SellerID	Score	Total Sales
01999		$41.311,56
01998		$41.264,61
01997		$46.972,23
01996		$42.444,51
01995		$57.334,59
01994		$53.156,04
01993		$42.034,04
01992		$39.409,23
01991		$43.047,82
01990		$32.612,28
01989		$51.512,94
01988		$55.102,09
01987		$51.411,79
01986		$48.868,53
01985		$48.771,49
01984		$46.709,57
01983		$49.798,47
01982		$39.807,53
01981		$53.172,01
01980		$39.423,66
01979		$43.405,98
01978		$48.068,09
01977		$57.928,44
01976		$37.088,51
01975		$46.850,45
01974		$41.502,88
01973		$44.491,61
01972		$51.925,86
01971		$69.657,82
01970		$49.275,26
01969		$49.504,38
01968		$56.728,10

Focused on individual customers, allowing searches by CustomerID, CustomerName, or SellerID.

Displays key metrics per customer, including total orders, total quantity, total spend, and average order value.

Includes a Customer Score card that aggregates quantity, orders, and total spend into a single performance index. Cards are color-coded: green for above-average customers, red for below-average.

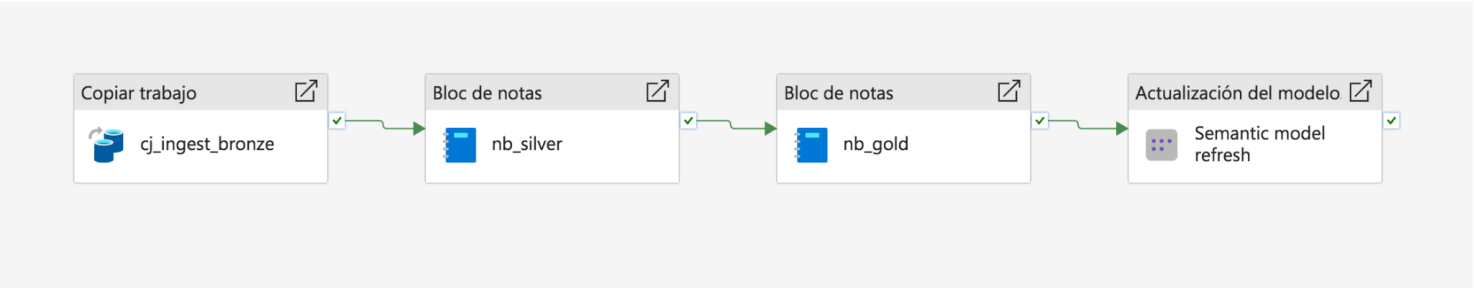
Tables include a score column, which combines quantity of products purchased, number of orders, and total spend to rank customers. This scoring logic is customizable based on client priorities (e.g., fewer orders with high revenue can be considered top-tier customers).

Interactive tables: Selecting one or multiple customers in the table immediately updates all report visuals, cards, and related KPIs. This allows users to analyze individual customer behavior or groups of customers in context, providing actionable insights for sales and account management.

Users can filter and drill through to see detailed transactions and behavioral patterns, supporting targeted analysis of high-value or risky customers.

Both reports include interactivity features such as tooltips, drillthroughs, filterable visuals, and table-driven filtering, allowing users to explore data at different granularities. These interactive elements make it possible to monitor trends, assess risk, and make data-driven decisions in real-time.

Automation and Pipeline



All processes (ingestion, Silver cleaning, Gold transformation, semantic model, and DAX measures) are automated in a pipeline that can run weekly to produce fresh, ready-to-analyze reports. This ensures reliable and up-to-date dashboards for business decision-making.